

Narrative Generation for Interactive Media: A Survey from Symbolic Systems to LLMs

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Abstract

This survey¹ traces the history of narrative generation as a subfield of natural language processing, from early symbolic systems to neural and LLM-based approaches. It argues that although the machinery has changed, the field repeatedly returns to the same core problems: coherence, character consistency, and the balance between narrative structure and creativity.

1 Introduction

Long before neural language models processed text algorithmically to generate narratives, artists used automatism to generate poetry. Dadaists used physical randomness: cutting up printed text, shuffling the pieces in a bag, and pulling them out to form new poetic lines. Surrealists used “automatic writing,” treating “free-wheeling human subjects” and their “unconscious drives” as a pseudorandom process to produce works such as *The Magnetic Fields* (1920) (Baillehache, 2013). Avant-garde artistic traditions form a prehistory for computational narrative generation: they used constraint, randomness, and automatism to produce narrative works, but not computation.

Narrative generation explores both human and computational narrative capacities, especially the cognition involved in storytelling. From early systems to contemporary LLMs, scholars repeatedly return to Aristotle’s *Poetics*, Dennett’s philosophy of character intentionality, and Grice’s Cooperative Principle to inform plot structure, believable agents, and user-system control, respectively. Decades after Dada and Surrealism, Nanette Wylde’s *Storyland* (2000) used computational randomization to generate text in pursuit of a similar effect: contemplation by the reader. Baillehache’s comparison of Dadaism, Surrealism, and Wylde’s *Storyland* bridges avant-garde and computational methods, highlighting their common processes and goals.

¹Word count: 2024.

2 The Symbolic Era

Narrative generation was formalized in computer algorithms in the mid-twentieth century, drawing from linguistics, literary studies, folklore, and early computer science to make narrative conventions computationally executable.

2.1 Early Symbolic Systems

Sheldon Klein’s 1971 murder-mystery generator, influenced by Roald Dahl’s “The Great Automatic Grammatizer,” was long treated as the earliest narrative generator until three earlier systems were identified (Ryan, 2017). These included SAGA II, which generated TV western scenes at MIT around 1960; Joseph Grimes’s 1961 Grimes’ Fairy Tales, which used Monte Carlo simulation and Vladimir Propp’s folktale templates during linguistic fieldwork in México; and Binnick’s 1968 Proppian fairy-tale generator, based on George Lakoff’s revision.

2.2 Symbolic AI Systems

In the symbolic AI era of the 1970s and 1980s, narrative generation drew on script-based narrative theories from Schank and Abelson, treating stories as combinations of scripts, plot templates, plans, agents, goals, and causal relations. Meehan’s TALE-SPIN was a landmark example: an interactive generator that produced Aesopian fables by generating stories as simulated agent behavior in a handcrafted world model (Meehan, 1977). Lehnert’s plot units encoded linked affect states into graphs for story summarization (Lehnert et al., 1981), while Lebowitz’s UNIVERSE used reusable plot “fragments” and a persistent cast of characters to generate serial soap-opera-style plot outlines, privileging author goals (maintaining drama) over character goals (happiness) (Lebowitz, 1985). Other systems explored learning: GENESIS generalized new story schemata from examples through

explanation-based learning (Mooney and DeJong, 1985), while MINSTREL used case-based reasoning and Transform-Recall-Adapt Methods to generate Arthurian tales by adapting solutions from prior story problems (Turner, 1993). These systems moved narrative generation toward richer models of plot structure, learning, and adaptation.

2.3 Interactive Narrative

Around the turn of the millennium, symbolic narrative research moved toward more interactive systems, emphasizing narrative intelligence, emergence within virtual environments, believable agents, drama management, and plot planning. The term “narrative intelligence” emerged to describe the human narrative faculty and to explore how it could inform AI research into generation (Mateas and Sengers, 1999). Meanwhile, work on “emergent narrative” explored the balance of control between user and system in interactive environments, arguing that increasing user agency could produce surprising and novel narratives (Aylett, 1999).

This turn toward interaction informed Façade, the first AI-based interactive drama. Façade created a playable real-time dramatic experience by combining hand-authored story material, believable agent characters, a drama manager, and an NLP system that parsed user input into “discourse acts” shaping character reactions. Its ABL architecture supported agentic character behavior, while the drama manager selected and sequenced authored story beats to maintain an Aristotelian story arc (Mateas and Stern, 2002, 2003). Façade marks an evolution of symbolic narrative AI into interactive drama: generation became less about open-ended text and more about balancing user agency, system flexibility, and coherent co-creation.

2.4 Planning

A related thread approached interactive narrative through AI planning. Interactive narrative could be modeled as a plan: a temporally ordered sequence of actions connected by causal links (Young, 1999). This framework managed plot coherence, but exposed a central design tension: if user action threatened the plan, the system could block it, preserving coherence at the cost of agency, or revise the narrative, preserving agency at the risk of incoherence. Planning was later extended by modeling narrative generation as a refinement-search problem, in which the system searches for character actions that transform an initial world state into a desired goal

state (Riedl and Young, 2010). The Intent-based Partial Order Causal Link planner, or IPOCL, adds character intentionality to causal planning by representing not only what characters do, but the goals that make their actions intelligible. In Dennett’s terms, IPOCL makes characters legible from the intentional stance: their actions matter because they advance the plot and can be understood as reasonable, goal-directed behavior.

3 Learned Narrative Models: From Event Chains to LLMs

By the late aughts, narrative understanding and generation shifted from authored story structures toward representations learned from text. Instead of hand-coding scripts, plot units, or planning rules, these systems explored how models could infer narrative patterns, track entities and events, and choose actions in worlds described in text. Yet this shift preserved older problems: coherence, consistency, world-state tracking, and user-system control.

3.1 Statistical Event Chains

Work on narrative event chains marked a shift from Schankian handcrafted scripts toward statistical models of narrative structure (Chambers and Jurafsky, 2008). Chambers and Jurafsky frame their system as the first attempt at automatic schema induction since GENESIS, but unlike earlier explanation-based approaches, their model induces narrative relations statistically from raw text. The system first uses an unsupervised distributional approach: coreference and typed dependencies identify event-role tuples associated with recurring participants, such as *hugged-object*, while pointwise mutual information estimates which tuples cohere into narrative chains around the same participant. A supervised temporal classification stage, trained on the TimeBank Corpus, imposes partial orderings on these induced chains. Although not itself a story generator, it bridges symbolic and neural approaches by learning character-centered plot structure from text rather than authoring it manually.

3.2 Neural Narrative Models

Neural narrative systems in the 2010s moved away from hand-authored symbolic knowledge while reintroducing symbolic-style structure to address the limits of purely sequential neural models. Agentic narrative understanding developed in the context of text-based gaming: one system used a Long

Short-Term Memory (LSTM) encoder to represent game text and infer underlying world states, while a neural Q-network scored which actions would maximize long-term reward, allowing the agent to interpret the world, act, and improve through interactive feedback (Narasimhan et al., 2015).

Other work addressed the limits of word-level recurrent story generation by introducing event representations as a middle layer between raw text and narrative structure (Martin et al., 2018). Martin et al.’s system preprocesses story text into abstract event representations, uses an event2event model to predict successor events, and then uses an event2sentence model to translate those events back into natural language. This preserves the symbolic intuition that stories are sequences of meaningful world-state changes, while replacing handcrafted plot knowledge with learned neural representations. A related method incorporated entity representations into a neural text generation model, drawing on Centering Theory to track characters as sources of narrative coherence (Clark et al., 2018).

Fan et al. address the broader challenge of long-form coherence: while recurrent and convolutional neural networks could model sentences, generating coherent long-form text required maintaining prompt relevance and long-range dependencies (Fan et al., 2018). Their hierarchical model generates a premise before the full story, while gated self-attention and model fusion help the generator remain conditioned on the prompt rather than drifting into generic language modeling. Taken together, these papers suggest that neural narrative generation did not simply replace symbolic narrative AI. It translated many of the same concerns — action, events, entities, world states, planning, and coherence — into learned representations and neural architectures.

3.3 Transformer-Based Language Models

In the early 2020s, narrative generation systems began addressing the same problems in the context of Transformer-based language models such as GPT-2. PlotMachines continues the pattern of augmenting neural generation with explicit narrative structure, using a dynamically updated representation of plot state to improve coherence (Rashkin et al., 2020). Building on the broader trajectory of prompt conditioning and entity tracking in earlier neural work, PlotMachines conditions on an outline of intended plot points and monitors which parts have already been realized in the generated story,

adding a memory mechanism to a Transformer-based architecture. Goldfarb-Tarrant et al. address global narrative coherence and quality by reintroducing content planning between the prompt and surface realization of a story: a plot-generation model proposes possible plots, which are reranked by rescoring models inspired by Aristotle’s *Poetics* (Goldfarb-Tarrant et al., 2020). Both systems show that controllable narrative generation depends not only on fluent natural language output, but on intermediate representations of plot progress, goals, and narrative constraints.

A parallel thread extends neural narrative understanding through reinforcement learning and text-based games. Building on the agentic framework of Narasimhan et al., CALM, or the Contextual Action Language Model, is trained on human game-play transcripts to generate compact sets of action candidates, which are then fed to a reinforcement-learning agent that reranks actions using game reward feedback (Yao et al., 2020).

3.4 LLM-Era (In)Coherence

Coherence remains a central problem in the LLM era. Transformer-based language models are the current state of the art in narrative generation, but still struggle with commonsense reasoning, producing plot incoherence over long narratives (Peng et al., 2022). Commonsense-inference Augmented neural StoryTelling (CAST) extends earlier attempts to improve coherence by conditioning generation on COMET inferences: commonsense predictions about causal links between events and character motivations. These inferences guide GPT-2 story continuations by supplying causal and motivational coherence through a neural commonsense module.

GROVE addresses LLMs’ difficulty generating stories with complex, credible plots. Rather than relying on increasingly detailed prompts, which can restrict creativity, GROVE retrieves similar human-written examples and uses an “asking-why” prompting strategy to build a “forest of evidence” around ambiguous or underdeveloped story details (Wen et al., 2023). By integrating useful evidence chains back into the draft, it uses retrieval, prompting, and revision to strengthen causality and produce more complex plots.

SWAG reframes narrative generation as a search for an “optimal path” through possible next actions (Pei et al., 2024). It uses two LLMs in tandem: one generates the next portion of a story, while a

second “action discriminator” evaluates the current story state and selects the best subsequent narrative action in a human-tunable, modular architecture. Together, these systems show that long-form narrative generation still requires guidance beyond local contexts: commonsense inference, retrieval, prompt engineering, iteration, and auxiliary LLMs.

3.5 Hybrid Futures

StoryVerse, a plot-generation prototype that balances authored structure with LLM-driven generation, provides a fitting endpoint for this survey, bringing together several threads that have shaped narrative generation from its inception (Wang et al., 2024). Like early symbolic systems, it remains concerned with author intent and coherent plot progression, but also allows for narrative emergence via flexible planning, neural generation, and player input, creating “living stories” co-produced by the author, LLM-driven system components, and the player. The author defines flexible story beats called “abstract acts,” while LLMs handle local narrative planning and agentic character simulation; the player can then take actions that meaningfully alter the plot while preserving the author’s broader story arc.

More recent work on SNAP (“Story and Narrative-based Agent with Planning”) extends this trajectory, showing how current LLM-based interactive narrative systems continue to depend on explicit structures for managing coherence, pacing, and disclosure (Bang et al., 2025). SNAP divides an interactive story into modular “Cells” governed by “Plans,” preserving author control while allowing local emergence in response to user input and preventing the model from breaking immersion by revealing information too early or drifting from the intended plot.

Recent systems like StoryVerse and SNAP therefore suggest a hybrid future for narrative generation, integrating handcrafted, authored structures with creative LLM-driven emergence.

4 Conclusion

Across its history, narrative generation has repeatedly changed its technical machinery while returning to the same problem: how to produce stories that are coherent, consistent, and artistic. Early symbolic systems formalized narratives; statistical and neural systems attempted to learn those structures from text; and LLM-era systems revisit

the same problems through commonsense inference, retrieval, and prompt engineering. The field’s evolution is therefore not a simple replacement of symbolic approaches by neural ones. Instead, it suggests that the future of narrative generation will likely depend on hybrid systems that combine the flexibility of neural models with the structure, intentionality, and controllability of symbolic systems.

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